

First Run Happy Path: Registration to Wallet Transfer

This document is for someone seeing the project for the first time. All normal client calls go through YARP Gateway.

Base URL

{{gateway}} = http://localhost:8080

FinWallet.Api on :8081 should not be called directly for normal client flows.

Example variables:

```
{{tokenA}} Customer A JWT
{{tokenB}} Customer B JWT
{{walletA}} Customer A TRY wallet ID
{{walletB}} Customer B TRY wallet ID
```

1. Register Customer A

POST {{gateway}}/api/v1/auth/register
Content-Type: application/json
X-Correlation-Id: demo-register-a

```
{
  "countryCode": "TR",
  "phoneNumber": "+905321111111",
  "email": "customer.a@example.test",
  "password": "Example-Password-A-123!"
}
```

Expected: HTTP 202 with registration pending response. Save customerId.

****OTP note:**** FinWallet never returns OTP in the registration response. FakeCommunication receives the raw OTP as the simulated SMS body. Read it only through local test/debug instrumentation; do not add a public OTP-leak endpoint.

2. Verify Customer A OTP

POST {{gateway}}/api/v1/auth/registration/verify
Content-Type: application/json

```
{
  "customerId": "<customer-a-id>",
  "code": "123456"
}
```

Expected: HTTP 200 and registration verified.

3. Login Customer A

POST {{gateway}}/api/v1/auth/login
Content-Type: application/json

```
{
  "phoneNumber": "+905321111111",
  "password": "Example-Password-A-123!",
  "deviceId": "demo-device-a"
}
```

Store response accessToken as tokenA. Never log the token.

4. Create Customer A TRY wallet

POST {{gateway}}/api/v1/wallets
Authorization: Bearer {{tokenA}}
Content-Type: application/json

```
{ "currency": "TRY" }
```

First call: HTTP 201 WALLET_CREATED; repeat for the same customer/currency: HTTP 200 WALLET_EXISTS. Store walletId as walletA. Initial available/blocked balances are zero.

5. Open Customer A external bank account

Not required for an internal wallet transfer but part of the intended onboarding story.

```
POST {{gateway}}/api/v1/bank-accounts
Authorization: Bearer {{tokenA}}
Content-Type: application/json

{ "walletId": "{{walletA}}" }
```

Provider pending may return HTTP 202. Retrying must not create a duplicate provider account because the durable BankAccount ID produces a deterministic provider request key.

6. Register + verify + login Customer B

Repeat steps 1-3 with different phone/email/device. Example phone: +905322222222. Store access token as tokenB.

7. Create Customer B TRY wallet

```
POST {{gateway}}/api/v1/wallets
Authorization: Bearer {{tokenB}}
Content-Type: application/json

{ "currency": "TRY" }
```

Store wallet ID as walletB.

8. Verify wallet list

```
GET {{gateway}}/api/v1/wallets
Authorization: Bearer {{tokenA}}
```

If JWT is missing/invalid, Gateway rejects the request before forwarding it to the backend.

9. Funding prerequisite — current gap

A new wallet starts at zero balance. A public BankDeposit/funding endpoint **does not exist yet**, so public APIs alone cannot complete register -> fund -> successful transfer.

Do not do this:

```
UPDATE Wallets SET AvailableBalance = ...
```

It creates money outside the Ledger and breaks reconciliation.

For transfer demos/tests, a controlled integration fixture must atomically create balance + FinancialTransaction + balanced LedgerJournal/Entries. BankDeposit should be the next real feature.

10. Wallet transfer

After walletA has been validly funded:

```
POST {{gateway}}/api/v1/transfers
Authorization: Bearer {{tokenA}}
Idempotency-Key: demo-transfer-a-to-b-0001
Content-Type: application/json
X-Correlation-Id: demo-transfer-a-to-b

{
  "sourceWalletId": "{{walletA}}",
  "destinationWalletId": "{{walletB}}",
  "amount": 125.50
}
```

Expected success: HTTP 200 WALLET_TRANSFER_COMPLETED, immutable transactionId, amount/currency/completedAt and wasReplay=false.

11. Safe replay

Send the same request with the same Idempotency-Key again. Expected:

- no second money movement;
- no second ledger posting;
- no second fraud evaluation for a completed replay;
- same transaction ID;
- WALLET_TRANSFER_REPLAYED;
- wasReplay=true.

Using the same key with a different amount/destination returns a conflict.

Gateway expectations

Request	Expected behavior
Register/verify/login/refresh without JWT	Gateway forwards
Wallet/bank-account/transfer without JWT	Gateway rejects
/providers/* without internal key	Gateway rejects
Backend business endpoint without downstream key	Destination rejects
Rate/body/header limits exceeded	Rejected before business processing

Swagger

```
http://localhost:8080/swagger Gateway
http://localhost:8081/swagger FinWallet.Api
http://localhost:8082/swagger FakeBank
http://localhost:8083/swagger FakeFraud
```

Normal business calls still go through Gateway :8080.